

MAT 549 HW02

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Problem 1. Compute the flow of each of the following vector fields on $\mathbb{R}^2$:

(a) $V = y \frac{\partial}{\partial x} + \frac{\partial}{\partial y}$

(b) $W = x \frac{\partial}{\partial x} + 2y \frac{\partial}{\partial y}$

(c) $X = x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y}$

(d) $Y = y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}$

Solution. (a) Let $\gamma_{(a,b)}(t) = (f(t), g(t))$ be the integral curve along V based at the point $(a, b) \in \mathbb{R}^2$, i.e. $\gamma_{(a,b)}(0) = (a, b)$. Then (dropping the subscript for brevity), γ satisfies

$$\gamma'(t) = f'(t) \frac{\partial}{\partial x} + g'(t) \frac{\partial}{\partial y} = V(f(t), g(t)) = g(t) \frac{\partial}{\partial x} + \frac{\partial}{\partial y}.$$

This differential equation has a solution over all of $\mathbb{R}$, namely $g(t) = t + b$ and $f(t) = \frac{1}{2}t^2 + bt + a$. The resulting flow is

$$\varphi_t^V(a, b) = \gamma_{(a,b)}(t) = \left(\frac{1}{2}t^2 + bt + a, t + b \right).$$

(b) Let $\gamma_{(a,b)}(t) = (f(t), g(t))$ be the integral curve along W based at the point $(a, b) \in \mathbb{R}^2$. Then, γ satisfies

$$\gamma'(t) = f'(t) \frac{\partial}{\partial x} + g'(t) \frac{\partial}{\partial y} = W(f(t), g(t)) = f(t) \frac{\partial}{\partial x} + 2g(t) \frac{\partial}{\partial y}.$$

We can solve the resulting differential equation to get $f(t) = ae^t$, $g(t) = be^{2t}$. The resulting flow is

$$\varphi_t^W(a, b) = \gamma_{(a,b)}(t) = (ae^t, be^{2t}).$$

(c) Let $\gamma_{(a,b)}(t) = (f(t), g(t))$ be the integral curve along X based at the point $(a, b) \in \mathbb{R}^2$. Then, γ satisfies

$$\gamma'(t) = f'(t) \frac{\partial}{\partial x} + g'(t) \frac{\partial}{\partial y} = X(f(t), g(t)) = f(t) \frac{\partial}{\partial x} - g(t) \frac{\partial}{\partial y}.$$

We can solve the resulting differential equation to get $f(t) = ae^t$, $g(t) = be^{-t}$. The resulting flow is

$$\varphi_t^X(a, b) = \gamma_{(a,b)}(t) = (ae^t, be^{-t}).$$

(d) Let $\gamma_{(a,b)}(t) = (f(t), g(t))$ be the integral curve along Y based at the point $(a, b) \in \mathbb{R}^2$. Then, γ satisfies

$$\gamma'(t) = f'(t) \frac{\partial}{\partial x} + g'(t) \frac{\partial}{\partial y} = Y(f(t), g(t)) = g(t) \frac{\partial}{\partial x} + f(t) \frac{\partial}{\partial y}.$$

We can solve the resulting differential equation to get $f(t) = \frac{a+b}{2} \cdot e^x + \frac{a-b}{2} \cdot e^{-x}$ and $g(t) = \frac{a+b}{2} \cdot e^x - \frac{a-b}{2} \cdot e^{-x}$. The resulting flow is

$$\varphi_t^Y(a, b) = \gamma_{(a,b)}(t) = \left(\frac{a+b}{2} \cdot e^x + \frac{a-b}{2} \cdot e^{-x}, \frac{a+b}{2} \cdot e^x - \frac{a-b}{2} \cdot e^{-x} \right).$$

□

Problem 2. For any integer $n \geq 1$, define a flow on the odd-dimensional sphere $S^{2n-1} \subseteq \mathbb{C}^n$ by $\theta(t, z) = e^{it}z$. Show that the infinitesimal generator of θ is a smooth nonvanishing vector field on S^{2n-1} .

Solution. The infinitesimal generator of $\theta(t, z)$ is the vector field $V(z) = \theta^{(z)'}(0) = iz$. Since no point on S^{2n-1} is zero, iz is also never zero, so this is indeed a nonvanishing smooth vector field on S^{2n-1} . $\square$

Problem 3. Suppose M is a smooth, compact manifold that admits a nowhere vanishing vector field. Show that there exists a smooth map $F: M \rightarrow M$ that is homotopic to the identity and has no fixed points.

Solution. We will show that for some $t > 0$ the flow $\varphi_t^V: M \rightarrow M$ of M along V satisfies the desired properties. Since the flow φ_0^V at $t = 0$ is the identity map, the flow at time t is homotopic to the identity for every t , so we only need to find a t for which φ_t^V has no fixed points.

Let $p \in M$ be an arbitrary point and $(x_1, \dots, x_m)$ a system of local coordinates on a neighborhood U of p such that the coordinates of p are $(0, \dots, 0)$ and $V(p) = \frac{\partial}{\partial x_1}$. We will alter the given coordinates to force $V \equiv \frac{\partial}{\partial x_1}$ in a neighborhood of p . Let Σ be the hypersurface given by the points inside U with $x_1 = 0$; clearly Σ is transverse to V at p , since $\frac{\partial}{\partial x_1}$ is not tangent to Σ .

Now, define $\Phi: (-\varepsilon, \varepsilon) \times \Sigma \rightarrow M$ by $\Phi(t, q) = \varphi_t^V(q)$, i.e. Φ is the flow of Σ along V ; $d\Phi_{(0,p)}$ is invertible because its derivative with respect to t at that point is just $V(p)$ and the differential with respect to q is $T_p\Sigma$ which, together with $V(p)$, spans T_pM . The upshot of this is that some neighborhood of $(0, p)$ gets mapped diffeomorphically to a neighborhood of p , so take a system of local coordinates $(y_1, \dots, y_m)$ on the former neighborhood with $y_1 = t$ and $(y_2, \dots, y_m) = q$ coordinates for Σ at p . Via Φ , these map to local coordinates $(\tilde{y}_1, \dots, \tilde{y}_m)$ on a neighborhood of p in which $V \equiv \frac{\partial}{\partial x_1}$, since flowing in the direction of V corresponds to moving in the y_1 direction.

What we have done is construct local coordinates in a neighborhood U_p of p for which flowing along V corresponds to translation, so there exists $t_p > 0$ such that φ_t^V has no fixed points in U_p for any $0 < t < t_p$. Since M is compact, there exist finitely many points $p_1, \dots, p_k \in M$ such that $U_{p_1} \cup \dots \cup U_{p_k} = M$, hence $\varphi_{t_0}^V$ has no fixed points where t_0 is chosen such that $0 < t_0 < \min_{1 \leq i \leq k} t_{p_i}$. $\square$